AnthracoForm

User guide — AnthracoForm v1.4 (CSV schema v1.4)

Supplementary material S2. Software: https://doi.org/10.5281/zenodo.20487673 (MIT licence). Data dictionary: S1.

This guide explains how to record wood-charcoal fragments with AnthracoForm, step by step. The screenshots show the Saharo-Sindian (Sindh) module with a demonstration file (site code DEMO); the Eastern Mediterranean (Lebanon) module works in the same way and differs only in its reference data. AnthracoForm is an aid to identification, not a substitute for it: reference atlases, comparison collections and time at the microscope remain the basis of every determination.

1. What you need

- Any up-to-date web browser. **Google Chrome or Microsoft Edge, version 86 or later**, are recommended: they let AnthracoForm read and write directly the CSV file you choose (File System Access API). In other browsers you open the file with **Open CSV** and save it with **Download CSV** (section 3.3).
- The HTML file of a regional module: AnthracoForm_Saharo-Sindian.html or AnthracoForm_Eastern-Mediterranean.html. Nothing else: no installation, no server, no account and no internet connection.
- A folder for your data, ideally one that is backed up, with one CSV file per site; optionally, a folder for the photographs (section 4.7).

1.1 Getting the application

1. Download the latest release from Zenodo (https://doi.org/10.5281/zenodo.20487673) or from the Releases page of the GitHub repository (ignaziominervini-ctrl/AnthracoForm).
2. Unzip the archive. Each module is in its own folder (Saharo-Sindian/, Eastern-Mediterranean/), with its README, CHANGELOG and LICENSE.
3. Copy the HTML file of your module to your computer and open it in a web browser (double-click, or drag it onto a browser window). Do not open it from inside the zip archive.

The file works offline: it includes everything it needs, including the libraries that produce the PDF record sheet.

2. The screen at a glance

At start-up a welcome window reminds you what the tool is for (Figure 1). Tick *don't show again on startup* to skip it next time.

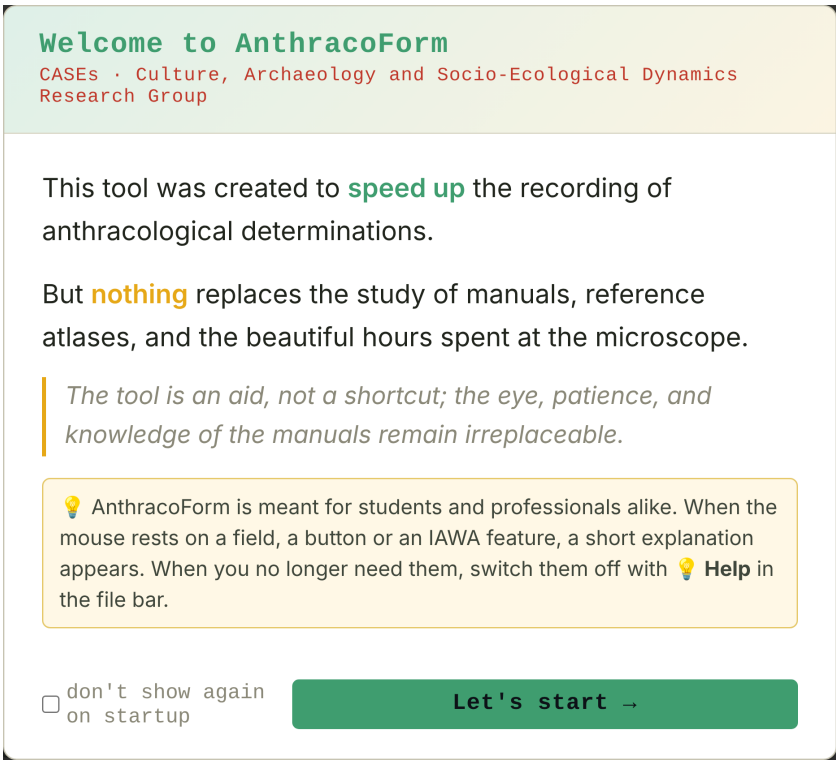


Figure 1. *The welcome window.*

The screen has three bands (Figure 2):

- **Header**: module name; record navigation (⇤ ← → ⇥); the position of the current record and the total number of records (*Tot*); ✓ **Save** (or ✓ **Update** when you are editing an existing record) and ✕ **Delete**. The badge on the left says whether you are writing a NEW RECORD or editing one (EDIT n/N).
- **File bar**: the file in use, time and operator of the last save and number of records; **Open CSV**, **New CSV**, ⟳ **Reload CSV** and 📷 **Photo folder** (section 4.7); **FIND**, to jump to a record by SU or by id_det; 💡 **Help**, which switches on or off the explanations shown when the mouse rests on any element of the form (on by default, meant for new users; the choice is remembered by the browser; section 2.1).
- **Tabs**: **Form** (recording), **IAWA Reverse Lookup**, **Dichotomous Key** and **Compare Taxa** (identification tools, section 6).



Figure 2. *Header, file bar and tabs, with the demonstration file open.*

2.1 Help for new users

AnthracoForm is meant for beginners (a student, an archaeologist with a bag of charcoal to be identified) as much as for people who have recorded charcoal for years. For beginners there is the 💡 **Help** mode: when the mouse rests on a field, a button, a channel, an IAWA feature or the saturation panel, a tooltip explains in one or two sentences what it is for and what it records (Figure 3). On IAWA features the tooltip gives the IAWA number and label and recalls how the feature is recorded: one click = present (absent in Absent mode), a second click = not assessed, D = diagnostic. The welcome window announces it (Figure 1).

Help mode is on at the first start. When the tooltips are no longer needed, a click on 💡 **Help on** in the file bar switches them off (the button becomes 💡 **Help off**); another click switches them on again. The browser remembers the choice for the following sessions. The tooltips change nothing in the data.

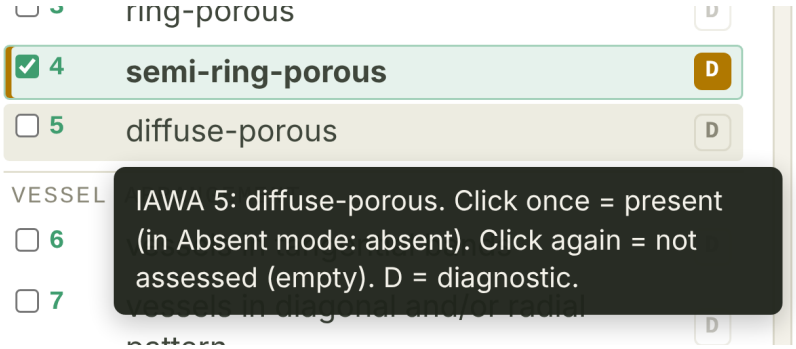


Figure 3. *Help mode: the tooltip on IAWA feature 5 (diffuse-porous). Above it, feature 4 is recorded as present and diagnostic (amber D).*

3. Starting a session

3.1 Opening or creating the site file

1. Click **Open CSV** and choose the CSV file of the site in the Windows (or system) file dialog. For a new site, click **New CSV**, choose the folder and type the file name. The dialogs reopen in the last folder used.
2. The browser asks for permission to edit the file: allow it.
3. The file bar shows the file name and the number of records loaded. The next free id_det is already filled in on the form.

From now on **every save is written to the CSV file on disk immediately**: there is no separate "save file" step, and the file on disk is always up to date. Use one file per site.

3.2 Reloading and changing file

- ⟳ **Reload CSV** reads the file again from disk (useful if you edited it elsewhere). Unsaved changes in the form are lost: AnthracoForm asks before reloading.
- **Open CSV** opens another file; **New CSV** creates a new one.

3.3 Browsers without folder access

In browsers without the File System Access API the file bar says *No folder access in this browser* and shows an **Open CSV** button. Click it to open an existing file, or start recording straight away for a new one. Records are kept in memory: save them with ↓ **Download CSV** (bottom right of the form), which downloads the file with the same name, and replace your copy with the downloaded one. Until you download, the file bar shows *changes not yet downloaded*, and the browser warns you if you try to close the page.

3.4 What the browser remembers

The data are only in the CSV file. The browser stores a few conveniences: the welcome-window setting, the fields locked with **KEEP**, the name of the last file, the list of taxa offered under the determination field, and the families and ecological groups you type for taxa that are not in the module's dictionary. These belong to the browser on that computer; clearing the browser data removes them, but never touches your CSV files.

4. Recording a fragment

The Form tab holds one fragment. Fill it in from top to bottom and save. Figure 4 shows the upper part of the form for a saved record.

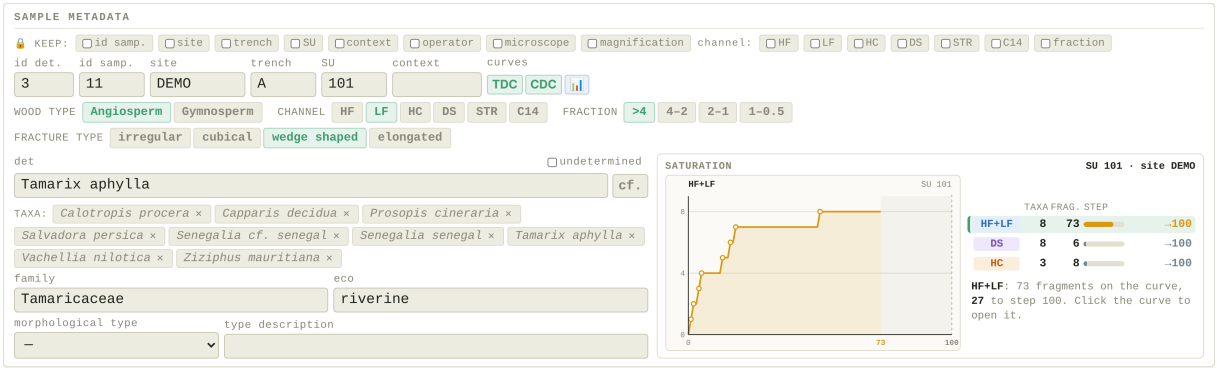


Figure 4. *Sample metadata, wood type, channel, fraction, fracture type, determination and saturation panel.*

4.1 Sample metadata

id det. is assigned automatically and gives the order of determination, which the curves follow. Fill in *id samp.* (sample or bag number), *site*, *trench*, *SU* and *context*. The SU, or the SU and context when the context is filled in, is the unit of the curves (section 7). Next to these fields, under *curves*, are the **TDC**, **CDC** and 📊 buttons.

KEEP (padlock row): tick a field to keep its value for the next records. Typically you lock site, sample, SU, operator, microscope and magnification once per SU and then record only the fragment. The same row lets you keep the **channel** and the **fraction**. You can tick several channels: the channel of a record passes to the next one if it is ticked, so with HF, LF, DS and HC ticked it always carries over; with only one channel ticked, every new record starts with that channel.

4.2 Wood type, channel, fraction, fracture type

Click a button to select a value; click it again to clear it.

- **Wood type**: *Angiosperm* or *Gymnosperm*. It decides which IAWA list is shown below (hardwood or softwood features).
- **Channel**: where the fragment comes from. The channel decides what the fragment can contribute to (Table 1); if your assemblage lacks some channels, see section 7.4.
- **Fraction**: >4, 4–2, 2–1 or 1–0.5 mm (stored as gt4, 4to2, 2to1, 1to05).
- **Fracture type**: irregular, cubical, wedge shaped or elongated.

Channel	Meaning	Curves and counts
HF / LF	Heavy and light fraction from flotation of a sample of known volume (typically 20 or 40 L).	Abundances and saturation curve of the SU.
DS	Dry sieving in the field, fixed mesh, volume not recorded.	List of taxa; conditional discovery curve (CDC).
HC	Hand-collected during excavation, selected by eye.	List of taxa; site-level taxonomic discovery curve

Channel	Meaning	Curves and counts
		(TDC).
STR	Closed context: hearth, oven, dump area, or a charcoal concentration distinct from the surrounding matrix.	List of taxa; analysed by event, never summed with the SU.
C14	Fragments selected for radiocarbon dating.	List of taxa only.

Table 1. *Recovery channels. Every determined fragment enters the list of taxa of its SU; only HF and LF enter abundances and the saturation curve.*

4.3 Determination

- Type the taxon in **det**. The **TAXA** buttons below list the taxa already recorded in the file: click one to insert it (the small ✕ removes it from the list, not from the data).
- **cf.** adds or removes "cf." in the determination. Fragments determined with cf., with a slash for several taxa (e.g. *Senegalia/Prosopis* type) or ticked as undetermined are recorded as usual but stay out of the curves: they count neither as fragments nor as taxa of a curve (Kabukcu and Chabal 2021).
- **undetermined**: tick it for fragments that cannot be identified.
- **family** and **eco** fill in automatically from the taxa dictionary of the module and can be edited. If you type family and eco for a taxon the dictionary does not know, AnthracoForm remembers them for that taxon in this browser.
- The **saturation panel** to the right of *eco* summarises the three curves for the SU and site of the current record (section 7.1). The fragments of an SU need not be recorded in a row: the curves follow the order of determination of all the records of the file.

4.4 Provisional morphological types

A fragment that you cannot identify yet can be assigned to a **morphological type** (tipo_1 to tipo_20 per file), with a short **type description**. When the type is identified, click **Resolve →**, enter det, family and eco and click **✓ Apply to all**: the determination is written to every fragment of that type. The type field of those fragments is cleared, but the type is not lost: it is kept in the tipo_resolved column, together with its description, and the form shows *resolved from tipo_n*.

4.5 Taphonomy, morphology, observed sections and photo status

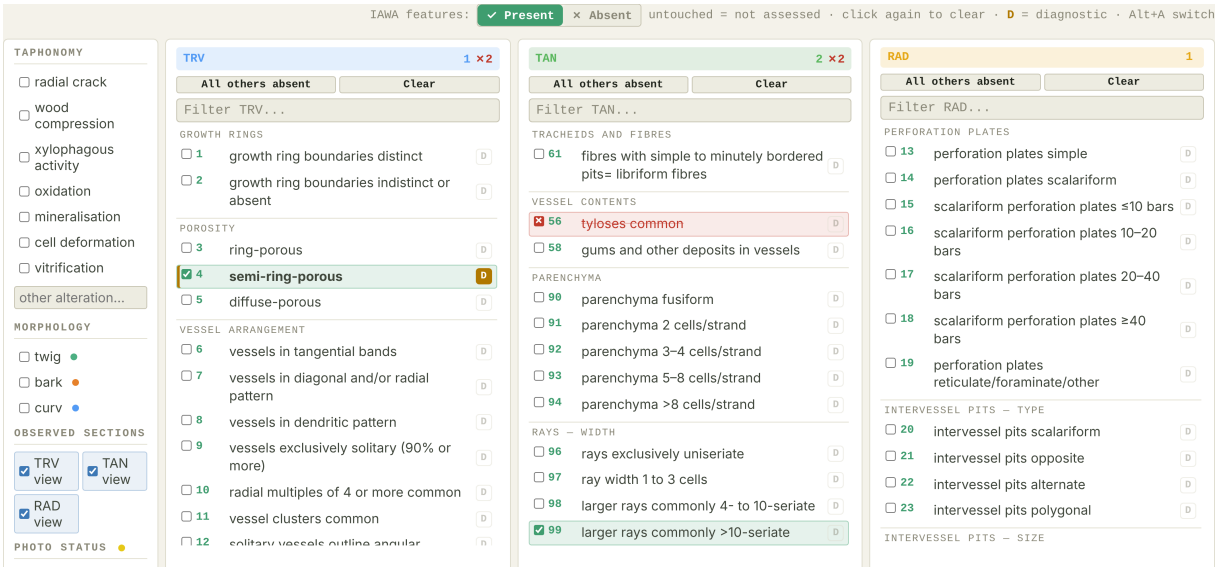


Figure 5. *Left: taphonomy, morphology, observed sections and photo status. Right: the IAWA checklists for the transverse (TRV), tangential (TAN) and radial (RAD) sections, with the present features in green and the absent ones in red; above them, the Present / Absent switch. The D box at the right of each feature marks it as diagnostic (amber: semi-ring-porous).*

- **Taphonomy**: tick the alterations present (radial cracks, wood compression, xylophagous activity, oxidation, mineralisation, cell deformation, vitrification). Each is recorded as 1 when seen; an alteration that is not seen is left empty, since it may be present but not visible on the fragment. An alteration not in the list goes in the **other alteration** field below, as free text (several separated by ";"); the alterations already recorded in the file are listed under it and added with a click.
- **Morphology**: *twig*, *bark*, and *curv* when the ring curvature can be measured to estimate the original diameter.
- **Observed sections**: tick TRV, TAN and RAD for the sections you actually examined. A feature can be recorded as absent only where its section was observed. The sections follow the features: clicking a feature ticks its section, and a section left without any recorded feature is unticked. Unticking a section by hand clears the features visible only in sections not observed.
- **Photo status**: *done* or *to do*.

4.6 IAWA features

The three panels list the IAWA features observable on a charcoal fracture surface, grouped by anatomical category: 102 hardwood features (IAWA Committee 1989) or 77 softwood features (IAWA Committee 2004), depending on the wood type. Each feature can be **present** (1), **absent** (0) or left **empty** (not assessed). The switch above the panels sets what a click records: in ✓ **Present** mode a click marks the feature as present (green), in ✗ **Absent** mode as absent (red, struck through); clicking a marked feature again clears it. **Alt+A** switches mode; the panels have a red border in Absent mode, and after each save the mode returns to Present. Features you do not touch stay empty: an absence is recorded only when you have looked for the feature. **All others absent** marks every untouched feature of the section as absent, for a complete description; **Clear** empties the section. The filter box above each panel searches by number or by text; the numbers at the top right of each panel count the present and the absent (✗) features. A feature visible in more than one section is shown in each of its panels and recorded once. The numbers on screen

are the IAWA feature numbers; in the CSV each feature is a column with the prefix of its list (e.g. iawa_A5, iawa_G40).

Diagnostic features. A feature is diagnostic only in relation to the taxa it separates: the same feature can decide one identification and be of no use in another. The small **D** box at the right of each feature marks the features that decided the identification of that fragment, not a fixed property of the feature: a present diagnostic feature is saved as **2** (required present), an absent diagnostic feature as **3** (required absent); the diagnostic box is highlighted in amber and the feature in bold. Clicking **D** on an untouched feature records it in the current mode (present or absent) and marks it as diagnostic; clicking D again removes the mark; clearing the feature removes it too. In the N_iawa boxes a diagnostic feature is typed with * (5*). Table 2 summarises the coding and how each value is entered.

Value in the CSV	Meaning	How it is recorded	On screen	InsideWood
empty	not assessed	feature not clicked, or cleared with a second click	not touched	—
0	absent: looked for and not seen, in an observed section	Absent, 1 click	red, struck through	a (absent)
1	present	Present, 1 click	green	p (present)
2	required present (present and diagnostic)	Present, 1 click + D	green, amber D	r (required present)
3	required absent (absent and diagnostic)	Absent, 1 click + D	red, amber D	e (required absent)

Table 2. *Coding of the IAWA features in AnthracoForm, how each value is entered and correspondence with the InsideWood search codes.*

Why four values. The coding follows that of the InsideWood searches (InsideWood 2004–; Wheeler 2011), where each feature of a description can be *present* (p), *absent* (a), *required present* (r) or *required absent* (e). In a search, p and a are tolerant criteria: they also let through the woods in which the feature is variable or unknown. r and e are strict criteria: they exclude every wood that does not have, or has, the feature. In AnthracoForm 1 corresponds to p, 0 to a, 2 to r and 3 to e, so a recorded hardwood fragment translates directly into an InsideWood search (for example 1p 5p 13r 104e).

Why absences and diagnostic features matter, on charcoal more than on wood. A recorded absence is as informative as a presence: knowing that multiseriate rays are missing excludes by itself every taxon that has them. On charcoal this weighs even more. Fragments are small, cracked, vitrified or distorted, and only a few features can usually be read on a fragment, often four to eight; those few must carry their proper weight. Keeping apart what was looked for and not seen (0) from what was not assessed (empty), and marking the few certain and decisive features (2 and 3), keeps in the file the reasoning behind the identification and lets anyone repeat it. As InsideWood warns, mark a feature as diagnostic only when you are sure: a wrong 2 or 3 excludes the right taxon. For multistate features (vessel diameter, ray width), marking one state present does not mean that the adjacent states are absent: use **All others absent** with judgement. In this version the diagnostic mark is recorded in the CSV and shown in the reports and profiles; IAWA Reverse Lookup and the Dichotomous Key treat the feature as an ordinary presence or absence.

4.7 Selected codes, photographs, notes and saving

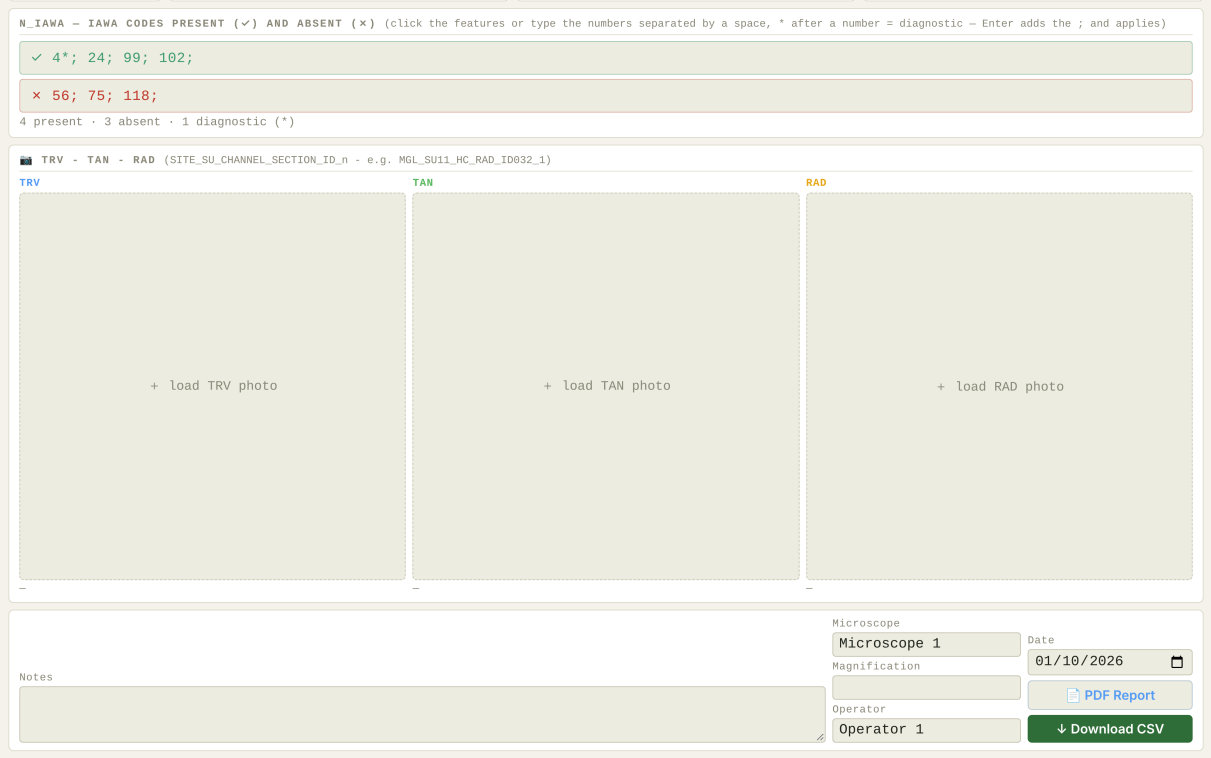


Figure 6. *Lower part of the form: list of present and absent IAWA codes, the photographs of each section, notes, operator, microscope, date, PDF record sheet and CSV download.*

- **N_iawa** lists the present (✓, green box) and absent (✗, red box) IAWA codes of the fragment. If you know the codes, you can also type them directly in the two boxes, just the numbers separated by a space (1 4 6; ; and commas work too) and press **Enter**, which adds the ; (or leave the box); a * after a number marks a diagnostic feature (5*): the features are marked in the checklists as if clicked, their section is ticked in Observed sections, and clicked and typed codes end up in the CSV in the same way. Deleting a number clears the feature; a code typed in both boxes counts in the box edited last; numbers are read in the list of the wood type chosen, and those not in it stay in the box, marked in red, and are not applied. No ranges (6-9): neighbouring IAWA numbers often exclude each other. **Esc** cancels.
- **Photos**: load the photographs of each section; a section can have several. The CSV keeps only their file names; the images stay in the photo folder of the project, always the same one, as in a GIS project. Choose it once with 📷 **Photo folder** in the file bar (Chrome or Edge): the browser remembers it and connects it automatically when the form is opened; the button only chooses it the first time or changes it. If the browser does not keep the permission (in Chrome/Edge you can choose "Allow on every visit"), it asks at the first click on the page. From then on the photos of each record are shown again whenever the record is opened. With the folder connected, ＋ **load** opens it directly and the names are saved relative to it, subfolders included; photos outside the folder are not added. Keep the folder and its subfolders where they are and with the same names: if the folder is moved or renamed, click 📷 and choose it again. Any file name can be used, but a laboratory standard that leads back to the record is recommended. Ours, also recalled in the title of the photo panel, is site, SU, channel, section, id_det and a progressive number: MGL_SU11_HC_RAD_ID032_1 is the first radial photograph of fragment 32, SU 11, hand collection. ✕ next to a name removes that photo from the record; the file is not deleted.

- **Notes** (several lines are allowed), **Microscope**, **Magnification** (the magnifications used, free text), **Operator** and **Date** (today's date by default; stored as YYYY-MM-DD).
- **✓ Save** (or **Ctrl+S**) writes the record to the CSV file and opens a new empty form with the kept values already in place. At each save the file is overwritten and keeps its name. The file bar shows the time of the last save and who made it (Last saved: 12:25 by Stefania); date and operator are in any case recorded in every record (date, operator). Writing directly to the file needs Chrome or Edge and the permission to edit the file, which the browser asks for when the file is opened: allow it, preferably with «Allow on every visit» (if it was refused, it is restored in the browser settings: Privacy and security, Site settings, File editing).

5. Finding and editing records

- Move through the records with ⇤ ← → ⇥ or **Alt+←/Alt+→** (from any tab).
- **FIND**: type an SU and press Enter to jump to its first record; press Enter again to cycle through the records of that SU. Type an id_det and press Enter to jump to that record.
- While a record is open the badge shows EDIT and the save button becomes **✓ Update**. Changes to an open record are saved automatically when you move to another record; a record you have not changed is never rewritten.
- **✕ Delete** removes the current record after a confirmation. It cannot be undone.

Keys	Action
Ctrl+S	Save or update the current record
Alt+← / Alt+→	Previous / next record
Alt+A	IAWA features: switch Present / Absent
Enter in FIND	Jump to the SU or id_det typed
Esc	Close the welcome window

Table 3. *Keyboard shortcuts.*

6. Identification tools

The three tools use the reference data of the module: the taxa with their IAWA profiles and the dichotomous key. They rank and compare candidates; the determination is always entered by the operator.

6.1 IAWA Reverse Lookup

The lookup uses the features ticked on the current fragment (shown at the top; the number on the tab counts them). It lists the taxa of the module ordered by the share of their reference profile that you have observed (*covered*) and, in case of a tie, by the number of matching features. For each taxon it shows the features that match, the features you ticked that the taxon does not have (*absent in taxon*), and the features of the taxon you have not ticked. Set a minimum number of matching features to hide weak candidates. **✓ Apply to det** copies the taxon into the determination field.

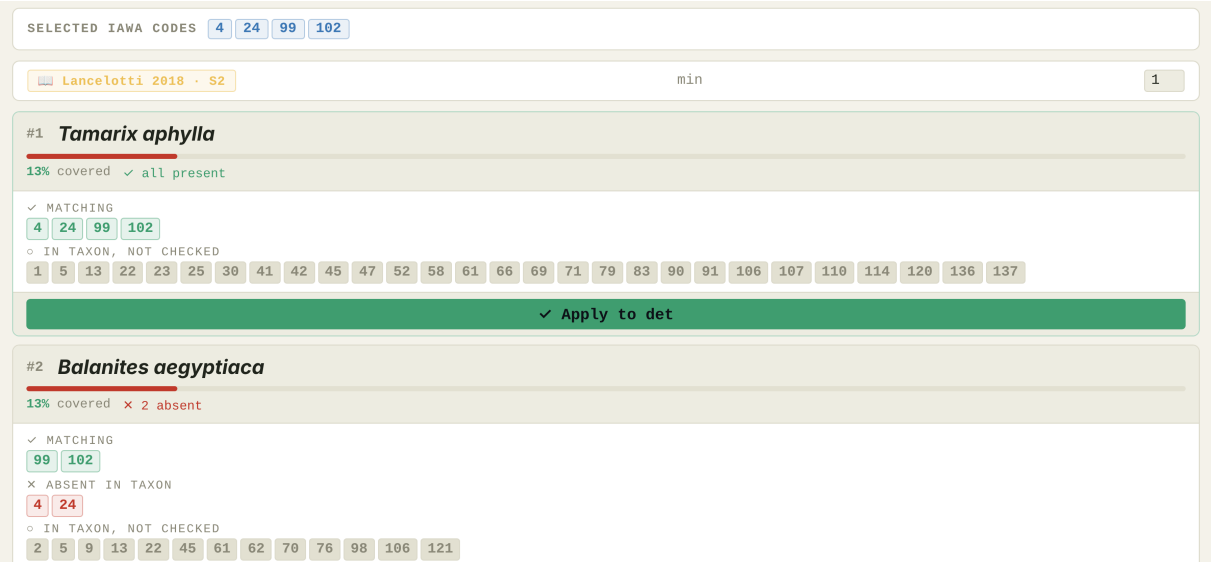


Figure 7. *IAWA Reverse Lookup for a fragment with four features ticked.*

6.2 Dichotomous Key

Follow the key one couplet at a time by clicking the statement that fits the fragment. The path taken is listed on the left; ← **Back** and ↻ **Restart** let you correct it. A result can be copied to the determination field. In the Saharo-Sindian (Sindh) module the key reproduces, couplet by couplet, the published key of Lancelotti (2018, Supplementary Material S2). In the Eastern Mediterranean (Lebanon) module it is the key to genera of Akkemik and Yaman (2012), and each result lists the species of that genus in the reference database; the transcription has been checked against the original pages (Akkemik and Yaman 2012: 293–296).

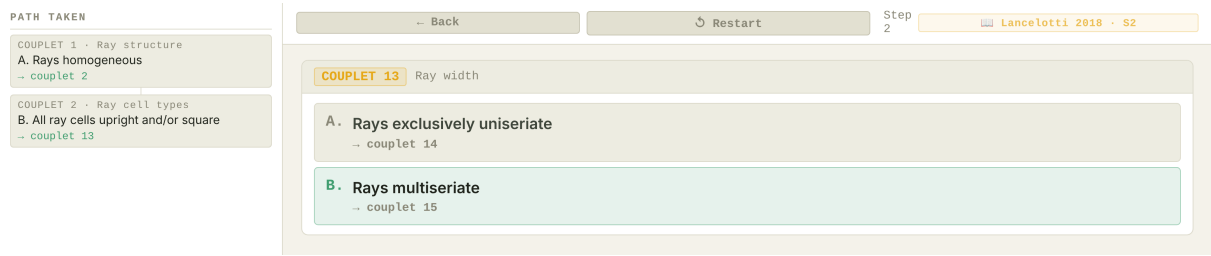


Figure 8. *Dichotomous Key after two couplets.*

6.3 Compare Taxa

Choose two or three taxa and click **Compare →**. The result lists the key distinguishing characters first, then, section by section, the features found only in one taxon and those shared.

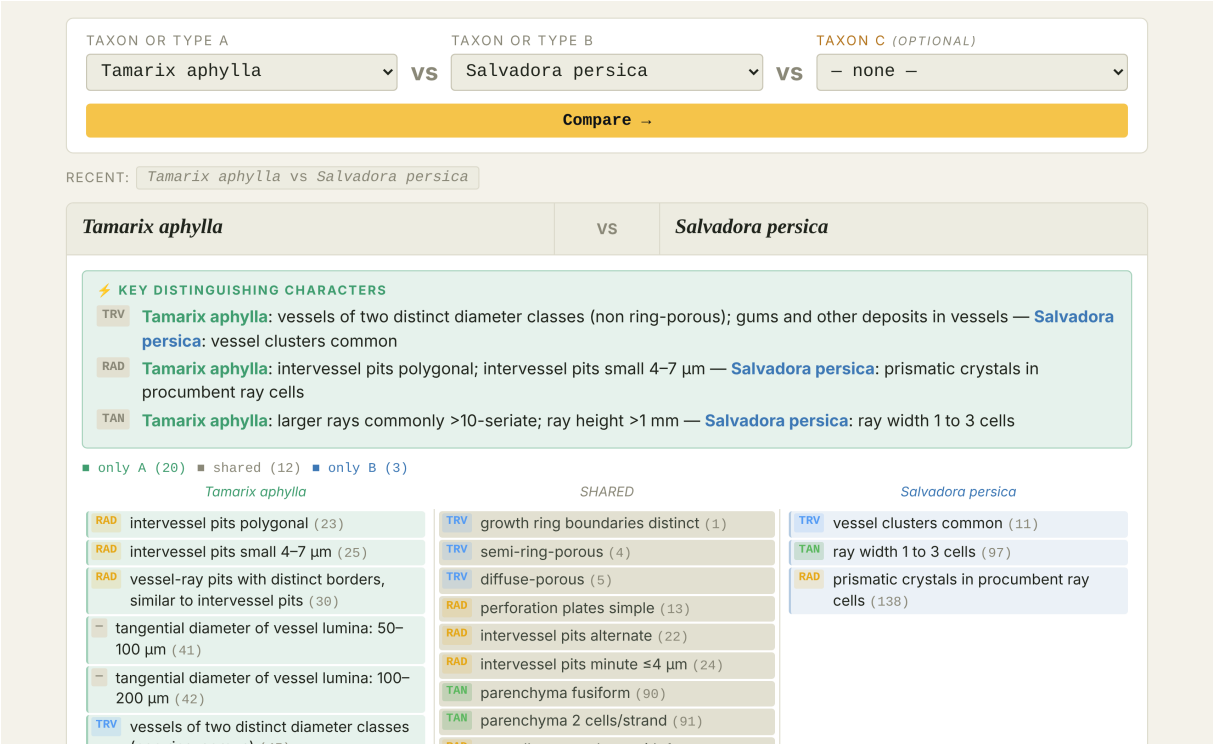


Figure 9. *Compare Taxa: two taxa side by side.*

The lists also offer the **provisional types of the open file** (section 4.4), so that a type not yet identified can be compared with the reference taxa or with another type. The profile of a type is calculated from its fragments each time the tab is opened: it includes the features present (1 or 2) in at least half of the fragments of the type in which they were assessed. It therefore grows with every fragment recorded and needs no manual updating.

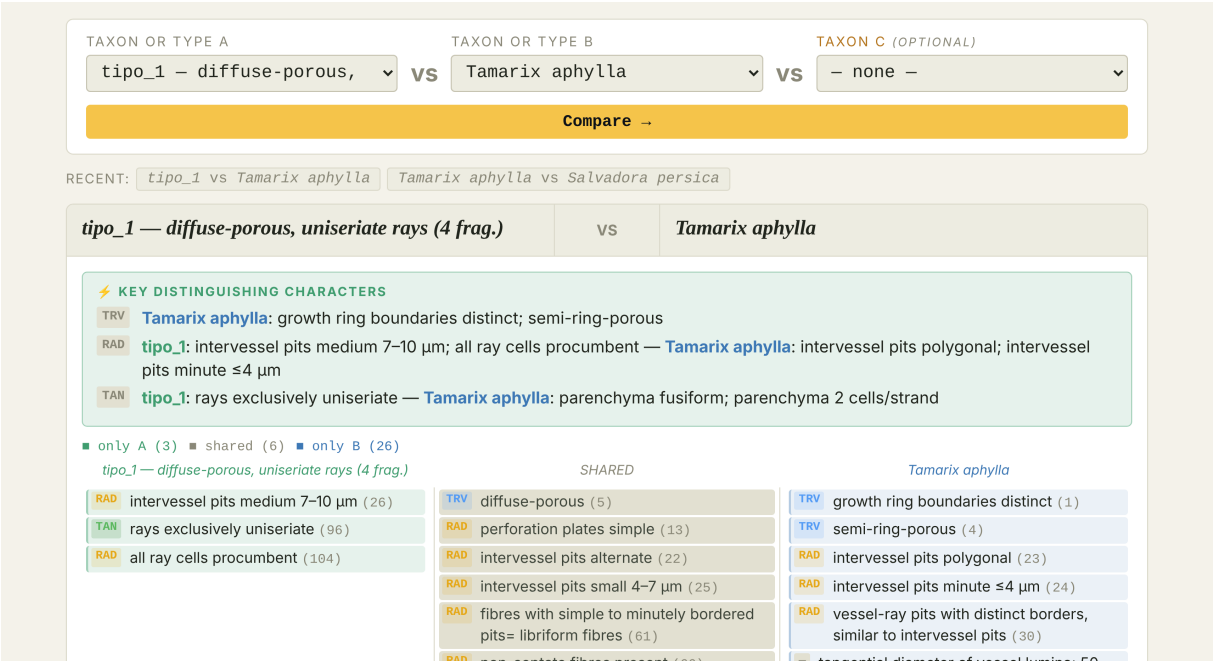


Figure 10. *Compare Taxa: a provisional type of the demonstration file (four fragments) compared with a reference taxon.*

7. Checking the sampling during the analysis

7.1 Saturation panel

A **saturation curve** (taxa-fragments curve) plots the identified fragments, in the order in which they were identified, against the cumulative number of taxa found. At first almost every fragment brings a new taxon and the curve rises steeply; then new taxa become rare and the curve flattens. When the curve stays flat over a long stretch, further fragments from the same unit add little to the list of taxa: this is how anthracologists judge whether a sample has been analysed enough (Kabukcu and Chabal 2021). AnthracoForm draws the curve while you record, so you can see where you are without leaving the form. On the small curve, look at the last stretch: if no open circles (new taxa) appear among the last fragments, the curve is flattening.

The panel to the right of the identification fields is always visible and changes with every record (Figure 11). On the left it draws the curve of the current fragment: the saturation curve of the unit (HF+LF), or the CDC (DS) or the TDC (HC) when the fragment comes from those channels. The horizontal axis reaches the next step, so the shaded part shows how many fragments are still missing; open circles mark new taxa and dashed lines the steps; a click opens the full-size curve. A line below the figures says how many fragments are on the curve and how many remain to the next step. On the right are the figures, one row per curve: **HF+LF** (saturation curve of the SU), **DS** (conditional discovery curve of the SU) and **HC** (taxonomic discovery curve of the site). The HF+LF and DS curves are computed per unit: the SU, or the SU and context when the context is filled in (SU 60 and SU 60 · context 1 are two curves, each with HF and LF together). The heading gives the unit and the site it refers to; when the SU field is empty, it refers to the last record saved, marked *(last record)*. The row of the channel selected in the form is highlighted.

Column	Content
taxa	Taxa on the curve. For DS, the count includes the taxa already found in HF and LF of the SU, as in the CDC.
frag.	Fragments on the curve (undetermined, cf. and slash determinations are not counted).
step	Next step for the fragments on the curve: 100, 200, 250, 300, 400 (e.g. →200); the last step reached is shown with a tick (✓100 …). The bar shows the way to the next step and turns from grey to yellow at 50 fragments and to green at 100.

Table 4. *Columns of the saturation panel.*

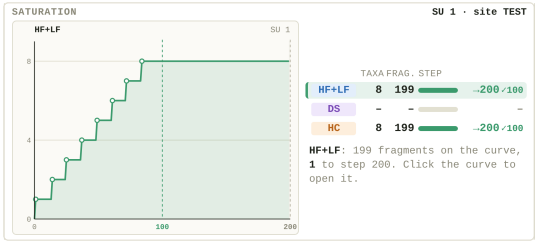


Figure 11. *Saturation panel on a test file: the HF+LF curve of SU 1 and the HC curve of the site have 199 fragments on the curve, past the 100 step and one fragment short of the 200 step; SU 1 has no DS fragments. The HF+LF row is highlighted because the current fragment is LF.*

The steps are reminders, not a stop rule. For assemblages from the arid zones of Sindh and Iranian Sistan, 100 identified fragments have proved to be a minimum; Kabukcu and Chabal (2021) give 200–300 identified fragments per layer for rich (e.g. tropical) assemblages and 250–

400 for temperate ones. The plateau of a curve has no mathematically defined end point: the decision to stop belongs to the analyst, and AnthracoForm never stops the recording.

7.2 Saturation curve and SU summary

The 📊 button (under *curves*) opens the summary of the current unit, SU or SU and context, or of the whole assemblage (Figure 12); the list at the top of the window lists the units as «60» or «60 · 1». The saturation curve and the CDC open in the same way. Its seven tabs are:

Tab	Content
1 · Saturation	Cumulative number of taxa against the HF and LF fragments on the curve, in order of determination. The bar above the chart gives the fragments on the curve, those left out (undetermined, cf., slash) and the steps reached; dashed lines mark the steps on the chart.
2 · Richness	HF and LF fragments per taxon. Taxa found in the SU only through other channels (DS, HC, STR, C14) are listed above the chart.
3 · Simpson	Simpson diversity (1 − D) of every SU, on HF and LF fragments, for comparison.
4 · Ecology	Composition of HF and LF fragments by the ecological groups defined in the module.
5 · Taphonomy	Share of the fragments of the SU on which each alteration was seen (Figure 13).
6 · Fracture	Distribution of fracture types.
7 · HF/LF	HF and LF fragments per SU.

Table 5. *Tabs of the SU summary.*

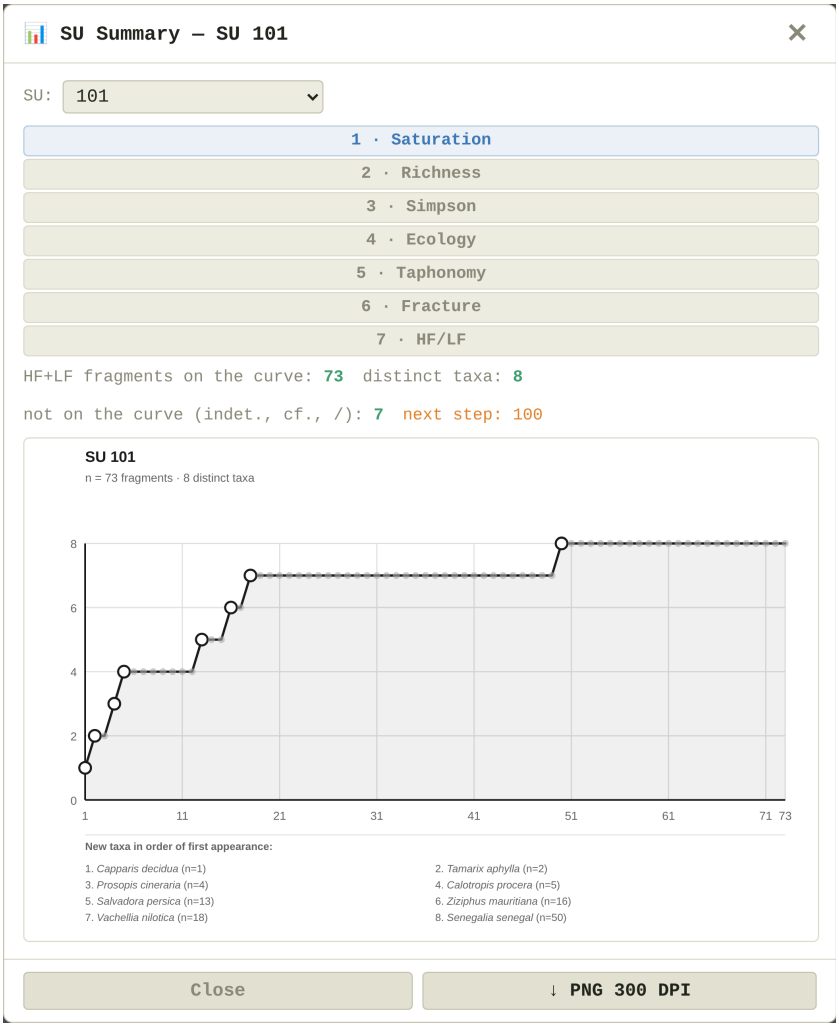


Figure 12. *SU summary, Saturation tab. Open circles mark the first fragment of each new taxon; the list under the chart gives the order in which taxa appeared.*

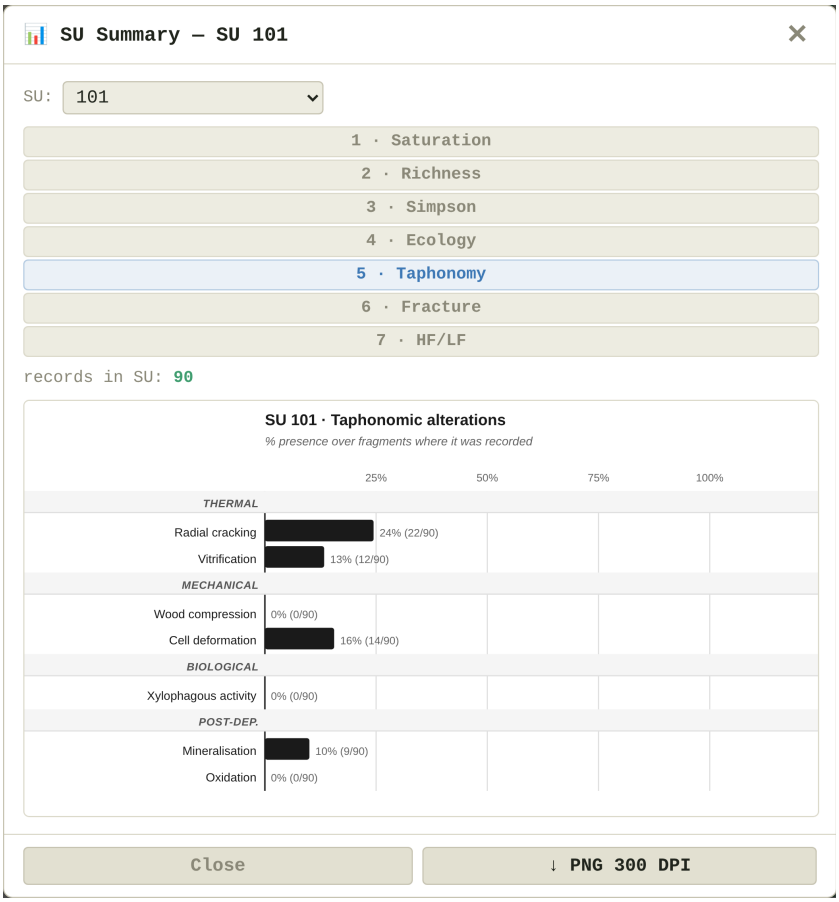


Figure 13. *SU summary, Taphonomy tab.*

Every chart can be exported with ↓ **PNG 300 DPI**: black and white, 170 mm wide, ready for print. These summaries are a check during the analysis; analyses for publication are run on the CSV file.

7.3 Curves for the other channels

- **TDC** (taxonomic discovery curve): one curve for the whole site, built on hand-collected (HC) fragments. It records how far the operator has learnt the local woods; it is not a saturation curve.
- **CDC** (conditional discovery curve): for each SU, built on dry-sieved (DS) fragments. It starts from the taxa already found in HF and LF and counts only the taxa that dry sieving adds; a flat curve means that DS adds nothing to flotation, which is itself a result.

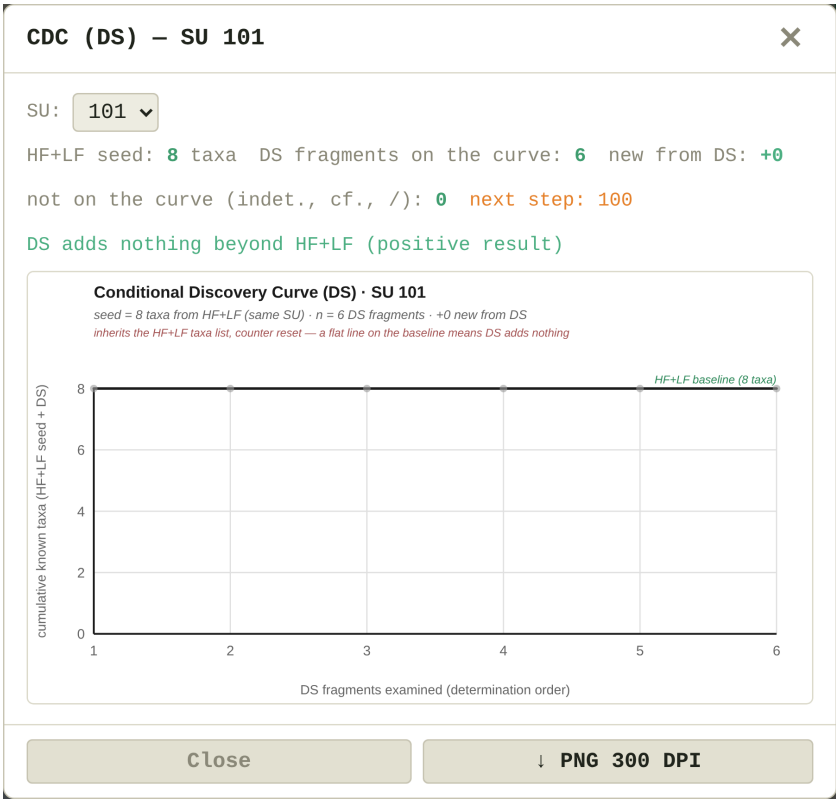


Figure 14. *Conditional discovery curve (DS) of an SU.*

Fragments from closed contexts (STR) and those selected for dating (C14) never enter the curves.

7.4 Using AnthracoForm with partial sampling

AnthracoForm was built on the complete sampling sequence used in the excavations of the ModAgrO and MeSAP projects: flotation of known volumes of sediment (HF and LF), dry sieving (DS), hand collection (HC), closed contexts (STR) and samples for dating (C14). Many assemblages have only some of these channels. The record is the same in every case; what changes is what the curves and the summary can tell (Table 6). Always record the channel the fragment really comes from.

Material available	Curves	SU summary (📊)	Read the data as
Complete sequence (HF, LF, DS, HC, STR, C14)	Saturation per SU (HF+LF); CDC per SU (DS); TDC per site (HC)	All tabs	Abundances and diversity from HF and LF; the CDC shows what dry sieving adds; the TDC records recognition
Flotation only (HF and/or LF)	Saturation per SU	All tabs	Abundances and diversity; with one fraction only, the curve is a curve of that fraction
Dry sieving only (DS)	CDC per SU, which without HF and LF is a plain saturation curve	Richness lists the taxa; Taphonomy and Fracture work; the abundance tabs stay	Relative frequencies calculated from the CSV file; no densities, because the volume is

Material available	Curves	SU summary (📊)	Read the data as
		empty	not recorded
Hand collection only (HC)	TDC per site only	Richness lists the taxa; Taphonomy and Fracture work; the abundance tabs stay empty	Presence of taxa and, with one id samp. per bag, ubiquity (share of bags in which each taxon occurs), calculated from the CSV file

Table 6. *What AnthracoForm offers with different sampling strategies. STR and C14 never enter the curves.*

Hand-collected charcoal is chosen by eye and is often concentrated: several fragments may come from one piece of wood, so a curve or a frequency counted on fragments says little about the vegetation or the fuel. Do not record hand-collected or dry-sieved fragments as HF or LF to obtain the curves: the file would no longer be comparable with those of other sites.

8. PDF reports

📄 **PDF Report**, at the bottom of the form, opens a small window with three choices. Every report is made of A4 pages that are always filled, is downloaded as a PDF file and works offline: the photographs take the space left by the text, and the text is reduced when it does not fit. The photographs come from the photo folder (section 4.7), which must be connected; to keep the PDF light they are reduced to 1400 pixels on the long side.

- **This record**: context, identification, taphonomy, morphology, observed sections, the IAWA features recorded in each section (✓ present, ✗ absent, ★ diagnostic, with their IAWA labels), notes and **all** the photographs of the fragment: up to nine on the first page, the others on the following pages, twelve per page (Figure 15). It is meant for doubtful fragments: a second operator can check it, it can be discussed with a specialist or filed with the reference material.
- **All fragments of a taxon**: a summary of the open file for that taxon: number of fragments, sites and SU, channels and fractions, taphonomy and morphology, operators, the IAWA profile (for each feature, the share of the fragments in which it was assessed that have it present, with the number of diagnostic records) and the list of the fragments; then **all** the photographs of the fragments of that taxon, six on the first page and twelve per page after that, each with fragment, SU and file name. It shows how a taxon looks on the charcoal of the site, and is a comparison sheet ready for the laboratory.
- **All fragments of a provisional type**: the same summary for a type (section 4.4), with the identifications already given to its fragments; a type not yet identified is titled *tipo_n — not yet identified* (Figure 16). It gathers the description, the profile and all the photographs of the type in one document: a way to describe it before identifying it and to show it to a specialist.

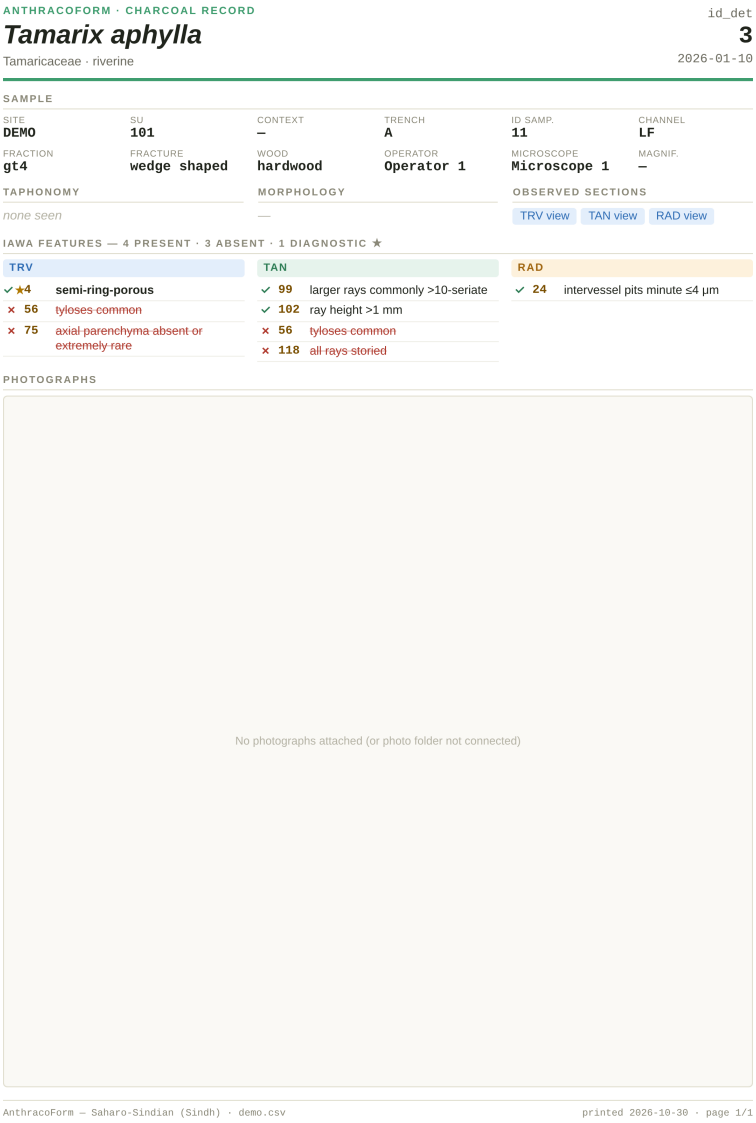
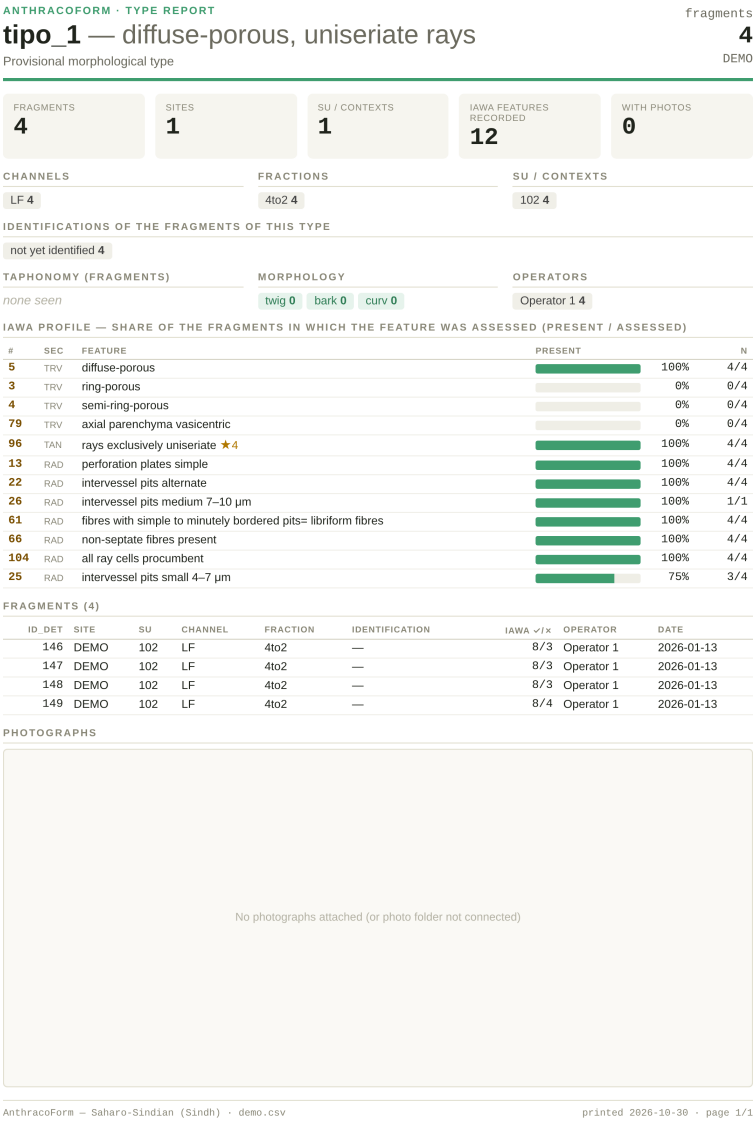


Figure 15. *PDF report of one record (demonstration file, without photographs).*



ANTHRACOFORM · TYPE REPORT

tipo_1 — diffuse-porous, uniseriate rays

Provisional morphological type

fragments 4 DEMO

FRAGMENTS	SITES	SU / CONTEXTS	IAWA FEATURES RECORDED	WITH PHOTOS
4	1	1	12	0

CHANNELS: LF 4

FRACTIONS: 4to2 4

SU / CONTEXTS: 102 4

IDENTIFICATIONS OF THE FRAGMENTS OF THIS TYPE: not yet identified 4

TAPHONOMY (FRAGMENTS): *none seen*

MORPHOLOGY: twig 0 · bark 0 · curv 0

OPERATORS: Operator 1 4

IAWA PROFILE — SHARE OF THE FRAGMENTS IN WHICH THE FEATURE WAS ASSESSED (PRESENT / ASSESSED)

#	SEC	FEATURE	PRESENT		N
5	TRV	diffuse-porous		100%	4/4
3	TRV	ring-porous		0%	0/4
4	TRV	semi-ring-porous		0%	0/4
79	TRV	axial parenchyma vasicentric		0%	0/4
96	TAN	rays exclusively uniseriate ★4		100%	4/4
13	RAD	perforation plates simple		100%	4/4
22	RAD	intervessel pits alternate		100%	4/4
26	RAD	intervessel pits medium 7–10 µm		100%	1/1
61	RAD	fibres with simple to minutely bordered pits= libriform fibres		100%	4/4
66	RAD	non-septate fibres present		100%	4/4
104	RAD	all ray cells procumbent		100%	4/4
25	RAD	intervessel pits small 4–7 µm		75%	3/4

FRAGMENTS (4)

ID_DET	SITE	SU	CHANNEL	FRACTION	IDENTIFICATION	IAWA ✓/✗	OPERATOR	DATE
146	DEMO	102	LF	4to2	—	8/3	Operator 1	2026-01-13
147	DEMO	102	LF	4to2	—	8/3	Operator 1	2026-01-13
148	DEMO	102	LF	4to2	—	8/3	Operator 1	2026-01-13
149	DEMO	102	LF	4to2	—	8/4	Operator 1	2026-01-13

PHOTOGRAPHS

No photographs attached (or photo folder not connected)

AnthracoForm — Saharo-Sindian (Sindh) · demo.csv

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Figure 16. *PDF report of a provisional type: its four fragments, their IAWA profile and the list of fragments (demonstration file, without photographs: with the photo folder connected, the photographs of the type fill the box and the following pages).*

9. The data file

Each row of the CSV file is a fragment; the 220 columns are the same in every module. The complete description is in the data dictionary (S1). The essentials:

- UTF-8, comma-separated; 1 = present or yes, 0 = absent or no; **an empty cell means "not recorded"**, never absent.
- IAWA columns: 1 = present, 0 = absent (looked for and not seen, in an observed section), 2 = required present (present and diagnostic), 3 = required absent (absent and diagnostic), empty = not assessed. The columns of the list that does not apply to the wood type are empty. The correspondence with InsideWood is in Table 2. Taphonomy columns: 1 = alteration seen, empty = not recorded.
- Dates in ISO format (YYYY-MM-DD); fractions coded gt4, 4to2, 2to1, 1to05 so that spreadsheets cannot turn them into dates.

- One file per site. Merge the files of a region only when the analyses are finished; site_code + id_det identifies each fragment in the merged data.
- The file can be read directly in R or in any statistical program.

Avoid saving the file from Excel. Excel may change the separator to a semicolon, alter dates or drop leading zeros. AnthracoForm reads semicolon-separated files, but the safest course is to open the file in Excel only for reading and to make changes in AnthracoForm.

9.1 Converting existing data

1. Start from the «Template» sheet of the data dictionary (S1), which has the 220 columns in the right order.
2. Copy your data into the matching columns, recoding values to the controlled vocabularies of S1 (channel, fraction, fracture type, 0/1). Leave empty what was not recorded; for IAWA features, write 0 only for features recorded as absent.
3. Give each fragment a progressive id_det. If the original order of determination is known, follow it: the curves depend on it.
4. Save as CSV (UTF-8), put the file in your data folder and open it with **Open CSV**. Check a few records on screen.

Files written with earlier versions of AnthracoForm (numeric IAWA columns without prefix, no wood_type, the column ossidation, no tipo_resolved) are converted automatically when opened and rewritten in the current schema at the first save.

From v1.4 a 0 in the IAWA columns means absent, and taphonomic alterations are 1 or empty. Files recorded with earlier versions, where 0 meant "not ticked" (absent or not assessed), must be corrected before use: replace with empty every 0 that is not a real absence.

10. Regional modules

Every module is the same program. The data that make it regional sit in one block at the top of the HTML file, between the lines REGIONAL MODULE DATA and END OF REGIONAL MODULE DATA. Everything outside that block is identical in all modules, so a correction to the program applies to every region. The block contains:

Element	Content
REGION	Module name and version; default wood type of a new record; the pair shown when Compare Taxa opens; the sources cited on screen and in the citation.
ECO_CATS	The ecological categories of the region, with the colour used in the charts and the rule that assigns an eco value to a category.
TAXA_DICT	Taxa dictionary: for each name, its family and ecological group (used to fill in family and eco).
TAXA_DB	Reference database: each taxon with the list of its IAWA codes (A or G prefix). Used by the Reverse Lookup and Compare Taxa.
KEY_DEF	The dichotomous key: each couplet with its statements, the next couplet or the result.

Table 7. *Contents of the regional block.*

Module names follow floristic regions (Loidi and Vynokurov 2024); the place in brackets is where the reference collection was made. Until v1.2 the modules were called Sindh and Lebanon.

To build a module for a new region, copy an existing HTML file, replace the content of the block with the data of the region, keeping the same structure, and leave the rest of the file untouched. Then open the file in a browser, walk the whole key and check a few taxa in the Reverse Lookup. The record structure, and therefore the CSV files, stay the same: data from different regions remain comparable.

Palms (Arecaceae) cannot be recorded with the current lists, because their stems have no secondary wood and are identified with features of their own. A third list of features, alongside those for gymnosperms and angiosperms, is in preparation.

11. Troubleshooting

Message or problem	What to do
"No folder access in this browser" in the file bar	The browser cannot write to a folder. Open and save the file with Open CSV and Download CSV (section 3.3), or open the HTML file in Chrome or Edge to save directly in the folder.
The browser asks again for permission to edit the folder	Normal at the start of each session: allow it.
"Data in memory — download CSV at end of session"	The file could not be written (for example because it is open in another program). Close the other program and save again, or download the CSV before closing the page.
"CSV with ";" detected"	The file was saved by a spreadsheet with European settings. It is read correctly and rewritten with commas at the next save.
The welcome window no longer appears	It was switched off with "don't show again on startup". Clearing the browser data brings it back.

Table 8. *Common messages and problems.*

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